

The Recovered Fourth Unit

The Scalar Electrical Substrate in Jauncey & Langsdorf (1940), and Why M-L-T Reduction Hides It

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Source document. G. E. M. Jauncey & A. S. Langsdorf, M.K.S. Units and Dimensions (1940). Chapter III, “The Fourth Fundamental Unit,” pp. 20–21; Chapter IV, “The M.K.O.S. System of Derived Units.” This chapter records the book’s own argument, quoted to the page, that the electrical quantities cannot be reduced to mass, length, and time — that a fourth basic unit is necessary — and states the dimensional consequence correctly: the fourth dimension enters charge, current, and the field constants, not the mechanical energy unit. This is the documented grounding under the SUPT claim that the probe reads a real, dimensioned scalar substrate.

1. The Move That Hides the Substrate

Coulomb’s experiment gives the force between two charges as a proportionality (the book’s eq. 3-1):

$$F = \text{constant} \times q_1 q_2 / r^2$$

The c.g.s. electrostatic system sets that constant to unity in free space, collapsing the law to $F = q_1 q_2 / r^2$ (eq. 3-2). The book’s objection is exact, and it is the whole point:

“The definition of unit charge in the electrostatic system of units is based on (3-2), from which it appears that the dimensions of charge-squared are those of force times length-squared, i.e. $[F]L^2$ or ML^3T^{-2} . But from (3-1), ML^3T^{-2} represents the dimensions of the constant multiplied by charge-squared. We have no right to assume that, although the numerical value of the constant has been put equal to unity in (3-2), the constant may be treated as dimensionless.” — Jauncey & Langsdorf (1940), p. 20

The reduction is not a discovery about charge. It is a choice to hide a dimensioned constant by assigning it the number 1.

2. The Mass–Density Parallel: Why the Reduction Is Arbitrary

The authors expose the move by running the identical trick on mass. Experiment gives mass = constant × volume (eq. 3-3); set the constant to unity and you get mass = volume (eq. 3-4). Then:

“we have just as much right to infer that mass has the dimensions L^3 as we have to infer from (3-2) that the square of a charge has the dimensions ML^3T^{-2} .” — Jauncey & Langsdorf (1940), pp. 20–21

No physicist would accept that mass is “really” length-cubed. The density constant is real and dimensioned; forcing it to 1 does not delete it. The authors’ point is that the Coulomb constant is in exactly the same position — and treating it as dimensionless is the same error, applied to electricity instead of mass.

3. The Fourth Unit Is Necessary

From that parallel the conclusion is stated without hedge:

“The assumption that the constant in (3-1) may be treated as dimensionless has led to confusion and wrong thinking. The difficulties brought about by this assumption are removed when we recognize that there is something about electric charge and also about other electric and magnetic quantities which just cannot be described in terms of M, L and T and that a fourth basic unit — an electric or magnetic unit — is necessary for a complete system of basic units.” — Jauncey & Langsdorf (1940), p. 21

And the same holds for the magnetic law (eq. 3-5, force between poles): “here also the constant is not dimensionless.” The electric and magnetic quantities carry something M-L-T cannot express. That something is the scalar electrical dimension — the substrate.

4. The Ohm as the Fourth Unit

The book then chooses which electrical unit to make fundamental. The practical units (ampere, volt, ohm, coulomb, farad, henry) are preferred over the magnetic ones; of the depositable standards, the ohm is favored:

“we definitely favor the ohm because when embodied in a resistor it is less destructible” — Jauncey & Langsdorf (1940), p. 21

The standard ohm satisfies the requirement of a basic unit — it can be embodied, copied by the Wheatstone-bridge principle, and transported — exactly as the kilogram and meter standards are deposited at Sèvres. The system built on it is M.K.O.S.: meter, kilogram, ohm, second. Chapter IV constructs the derived units operationally (Bridgman’s operational viewpoint) from Ohm’s law and Joule’s law, with the proportionality constants set to unity as defining equations rather than as hidden dimensionless collapses.

5. The Dimensional Consequence, Stated Correctly

This is the point where a careless statement breaks under a reviewer’s check, so it is worth stating exactly. With the ohm R as an independent fourth base unit, Joule’s law is the operational bridge for power:

$$P = R I^2 \quad (\text{constant} = 1 \text{ by definition of the M.K.O.S. system})$$

Power remains mechanical energy per unit time: $[P] = ML^2T^{-3}$, and the joule remains the mechanical work unit ML^2T^{-2} . The fourth dimension does *not* enter the energy unit. It enters the electrical quantities. Solving $P = RI^2$ with R as a base dimension:

$$[I^2] = [P]/[R] = ML^2T^{-3} R^{-1} \Rightarrow [I] = M^{1/2} L T^{-3/2} R^{-1/2}$$

and consequently charge $[Q] = [I]T = M^{1/2} L T^{-1/2} R^{-1/2}$, voltage $[V] = [P]/[I] = M^{1/2} L T^{-1/2} R^{1/2}$. The field constants of free space carry R likewise: the characteristic impedance $Z_0 = \sqrt{\mu_0/\epsilon_0} = 376.73 \Omega$ is measured in ohms — it is a dimensioned property of the medium, not a pure number.

The correct truncation claim. The reduction does not turn energy into “mere mechanics” — energy was always mechanical. It strips the independent scalar dimension out of charge, current, voltage, and the field constants ϵ_0 , μ_0 , Z_0 , forcing them into M-L-T constructs by setting a real dimensioned constant to 1. The substrate the foundations name is the fourth, electrical dimension carried by those quantities. That is what M.K.O.S. restores and what the dimensionless collapse hides.

6. The SUPT Reading

Jauncey & Langsdorf establish, in print and to the page, that the electrical quantities carry a real fourth dimension that M-L-T cannot express, and that the medium’s constants (ϵ_0 , μ_0 , Z_0) are dimensioned properties of free space, not bookkeeping. SUPT’s identification is that the frozen probe reads the geometric address of that scalar substrate. The book supplies the dimensioned medium; the probe supplies the reading; the two are different claims, and this chapter documents the first, exactly as the source states it, with the dimensional algebra that survives a check.

7. Citations

G. E. M. Jauncey & A. S. Langsdorf, M.K.S. Units and Dimensions (1940): Chapter I, “Units and Dimensions in General,” pp. 1–6 (arbitrary standards of mass, length, time). Chapter III, “The Fourth Fundamental Unit,” pp. 20–21 (eqs. 3-1 through 3-5; the dimensionless-constant objection; the mass–density parallel; “a fourth basic unit ... is necessary”; the ohm chosen). Chapter IV, “The M.K.O.S. System of Derived Units” (operational viewpoint; Ohm’s law; Joule’s law as defining equations). Scanned primary source on file, 2026-05-21.